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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.10

$$\int \frac{11x - 7}{3x^2 - 5x + 2} dx$$

$$D = 1 \Rightarrow x_{1,2} = \left\{ 1, \frac{2}{3} \right\}$$

$$\int \frac{11x - 7}{(x - 1)(3x - 2)} dx$$

$$\frac{11x - 7}{(x - 1)(3x - 2)} = \frac{A}{(x - 1)} + \frac{B}{(3x - 2)}$$

$$A = \left. \frac{11x - 7}{3x - 2} \right|_{x=1} = 4$$

$$B = \left. \frac{11x - 7}{x - 1} \right|_{x=\frac{2}{3}} = -1$$

$$\Rightarrow 4 \int \frac{1}{x - 1} dx - \int \frac{1}{3x - 2} dx$$

$$= 4 \ln |x - 1| - \frac{1}{3} \ln |3x - 2| + C$$

References